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and sensors for direction, temperature and acidity measurements. The body is made up of anodised aluminium, which is rendered waterproof by coating it with hard chromium.

The bump sensor lets the robot know it has collided with something and this is communicated to the user. The digital pressure gauge determines the pressure on the robot's body. If the pressure underwater exceeds a certain value, it chokes. This information shows up on the user's laptop so that the robot can be pulled up.

Once SaUsR reaches the bottom of the tank, the counter suction unit gets into action. It has two brushes in a cavity that rotate in different directions using coreless DC motors with very high-precision gearing mechanism. This scrubs the sludge 'off' and creates a whirlpool of water and the sludge. The pumping unit quickly sucks out the mixture of water and sludge.

The user can choose from two pumping mechanisms. In one method, the unit sucks out the contaminated water and disposes it of using an outlet pipe. In the other method, a filter attached to the robot separates the sludge deposit from the contaminated mixture while it is still under the cavity.

Gridbots has applied for patents for the underwater actuator design, the cleaning system design and the control system of the actuators.

Hurdles along the way

Keeping the weight of the device low and ensuring minimal power consumption was a big challenge.

Gaur shares, "We developed our own device using minimal hardware and gave the system the power to self-lock, that is, components go into sleep mode when they are not in use. For instance, when the camera is not in use, it goes blank and consumes much less power than it usually does."

"Availability of equipment such as underwater actuators was a challenge as they are quite expensive. So we designed and manufactured parts such as underwater mechanical drives and active pressure measuring system by ourselves, which turned out to be economical. Also, we used aluminium instead of iron to reduce SaUsR's weight," he adds.

Future plans

Some chemical companies use water as raw material in their products and store it in tanks the size of buildings. The temperature of some of the water tanks is regulated at 3-4 degrees celsius. Gridbots team is designing a robot that can work in such environments. They intend to target industries and government establishments with the new version. ●

The author was business correspondent at EFY Bengaluru